

Toward an improved lower bound for the largest prime factor of an odd perfect number

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Abstract

Goto and Ohno proved in 2008 that every odd perfect number has a prime factor exceeding 10^8 (100 million) [4]. A 2013 conference abstract by Mori, Sawae, Aoki, and Ishii announces the stronger threshold 10^9 (1 billion) [6]; a full proof of that announced improvement was not located in our literature search. We describe and audit a computer-assisted approach to the proposed assertion that every odd perfect number has a prime factor exceeding $B = 4,294,967,291$. The mathematical contribution of this draft is an explicit reduction to finite cyclotomic completeness and rooted-hypergraph obligations, with proofs of the reduction and of the conservative pruning rules. Historical Rust, C++, and GPU computations report that these obligations hold. The preserved archive contains 3,299 prime-index checkpoints, 12,553 positive cyclotomic factorizations, and 100 high-index enumeration shards. An additional validator rechecks every stored positive factorization by exact arithmetic and trial-division primality tests. The audit nevertheless identifies an unresolved 32-bit GPU hit-counter overflow obligation and a distinction between aggregate execution records and independently replayable certificates of exhaustive rejection. Consequently this draft does *not* establish the proposed bound unconditionally and does not claim an accepted record. It supplies a reviewable conditional argument, the available evidence, exact reproduction instructions, and the remaining verification work. This revision also gives a quantitative whole-system edge/index alternative, a stronger localized condition for a powerful Euler half-index, a restricted frontier result, and a conditional application of an existing unequal-index gcd theorem. These later arguments preserve arbitrary support and full valuations but leave explicit universal gaps. The existence of odd perfect numbers is not resolved.

1 Introduction and contribution

For a positive integer n , let $\sigma(n) = \sum_{d|n} d$. An odd perfect number would be an odd integer $N > 1$ satisfying $\sigma(N) = 2N$. Write $P(N)$ for its largest prime factor. The candidate statement examined here is

$$P(N) > 4,294,967,291. \tag{1}$$

This is a universal restriction on hypothetical odd perfect numbers, not the construction of a large prime, nor a proof that no odd perfect number exists.

The completed research was audited from its source and saved artifacts, not from the historical files' status labels. Those files include the string `complete_world_record_proof`; that string is a software output label and is not adopted as the conclusion of this paper. Our principal established mathematical statement is the conditional implication in Theorem 4.5. Application at B requires the finite obligations specified below. Their historical outputs agree, but important verification obligations remain open (Section 7).

The focused contribution is a self-contained account of prime-index coverage, an exact finite reduction of the unrestricted exponent problem, and a root-label method for simultaneously overapproximating possible supports of all Euler primes. A new pure-Python implementation checks positive witnesses and regenerates a complete small-bound example using a different algorithm. We make no general priority claim for elementary lemmas or fixed-point techniques. Secondary conditional results and failed approaches are separated in the supplement rather than treated as evidence for (1).

The numerical audit and its reruns date from 26 August 2026. Section 6 incorporates selected later full-system work through 2 September, with a missing review completed and the rewritten arguments checked on 3 September. That work is logically separate from the candidate-bound computation: it neither repairs the outstanding computational obligations nor asserts a global nonexistence proof. No new large arithmetic search was run for this revision.

1.1 Related work and priority status

Goto and Ohno proved $P(N) > 10^8$ in *Mathematics of Computation* [4]. Their author-maintained account describes a cyclotomic improvement of the Hagis–Cohen approach and use of earlier Jenkins programs [3]. In the official 2013 Mathematical Society of Japan meeting abstracts, Mori, Sawae, Aoki, and Ishii announce $P(N) > 10^9$ and describe their method as an improvement of the Goto–Ohno method [6]. The retrieved abstract is an announcement, not the full proof or reproducible computation. It must be acknowledged in any priority comparison; it should not be silently discarded merely because a full paper was not located.

A targeted literature search was conducted on 26 August 2026 using the exact paper titles, the four conference authors, and variants of “largest prime factor/divisor” with “odd perfect number”, including 2025–2026 results. No subsequent fully documented improvement of this particular lower bound was located in that search. This is not proof of absence of such work. For context, Zelinsky’s 2021 paper still uses the 10^8 bound and discusses the difficulty of reaching 10^9 [9, pp. 42–43]; that does not invalidate the earlier conference announcement.

The author project page, publisher-deposited Crossref metadata and abstract, and the official 2013 abstract were checked. Direct retrieval of the full Goto–Ohno journal PDF and extended author PDF failed during this audit; a fresh line-by-line comparison of those complete proofs is therefore not claimed. The access record and remaining literature checks are in LITERATURE.md. If established, (1) would exceed both identified thresholds (by factors approximately 42.95 and 4.295, respectively). Neither that arithmetic comparison nor the present search establishes an accepted world record or external validation.

2 Definitions and elementary reductions

Definition 2.1. For fixed $B \geq 3$, a pair (p, r) is *acceptable* if $p \leq B$ is an odd prime, r is an odd prime, and every prime divisor of $\Phi_r(p) = 1 + p + \cdots + p^{r-1}$ is at most B . Its dependency set is $D(p, r) = \text{supp}(\Phi_r(p))$, with multiplicities discarded only after acceptability has been established.

Lemma 2.2 (Euler form and necessary dependencies). *An odd perfect number has the form $N = q^a \prod_{p \neq q} p^{a_p}$, where $q \equiv a \equiv 1 \pmod{4}$ and every a_p is positive and even. Every ordinary prime p has an odd prime $r \mid a_p + 1$, and $\Phi_r(p) \mid \sigma(p^{a_p})$. If $a > 1$, the same assertion holds for q with an odd prime divisor of $a + 1$. If $a = 1$, the odd dependencies of the Euler component are exactly $\text{supp}((q + 1)/2)$.*

Proof. Multiplicativity gives $\sigma(N) = \prod_p \sigma(p^{a_p})$ and $\nu_2(\sigma(N)) = 1$. For odd p , $\sigma(p^b)$ is odd exactly when b is even. Thus exactly one exponent is odd. For that component the elementary

2-adic lifting identity gives $\nu_2(\sigma(q^a)) = \nu_2(q+1) + \nu_2(a+1) - 1 = 1$. Both terms are positive, so both equal one, proving the congruences. The identity $\sigma(p^b) = \prod_{d|b+1, d>1} \Phi_d(p)$ proves the divisibilities. If $a > 1$ and $a \equiv 1 \pmod{4}$, then $(a+1)/2$ is an odd integer greater than one. Finally $\sigma(q) = q+1$, whose 2-adic valuation is one. Every odd prime factor of any component sum divides N , since it divides $\sigma(N) = 2N$. $\square$

Lemma 2.3 (Factor classification and complete index ceiling). *For odd primes p, r , a prime $\ell \mid \Phi_r(p)$ is either $\ell = r$, in which case $p \equiv 1 \pmod{r}$ and $\nu_r(\Phi_r(p)) = 1$, or $\text{ord}_\ell(p) = r$ and $\ell \equiv 1 \pmod{2r}$. Every acceptable pair has $r \leq \lfloor (B-1)/2 \rfloor$.*

Proof. The value is odd and is 1 modulo p . If $\ell \neq r$ divides it, then $p^r \equiv 1 \pmod{\ell}$. If $p \equiv 1 \pmod{\ell}$ the cyclotomic value is r modulo ℓ , impossible. The order is therefore r and divides $\ell-1$; oddness gives divisibility by $2r$. For $\ell = r$, the value is nonzero modulo r unless $p \equiv 1 \pmod{r}$; in that case the lifting identity $\nu_r(p^r - 1) = \nu_r(p - 1) + 1$ proves the asserted multiplicity. Since $\Phi_r(p) > r$, it cannot consist only of this one ramified factor. There is a nonramified factor $\ell \geq 2r + 1$, proving the ceiling. This argument needs no unexamined primitive-divisor exception. $\square$

Only a necessary prime-index divisor is retained from each full exponent. There is no assertion that the full exponent index is prime, that all ordinary exponents equal two, or that a prime-index hyperedge accounts for every divisor of a composite-index component. Relaxing these additional conditions can introduce false candidates but cannot remove a true support.

3 Cyclotomic completeness obligations

Set

$$Q_B(r) = \prod_{\substack{\ell \leq B \\ \ell \equiv 1 \pmod{2r}}} \ell.$$

Lemma 3.1 (Exact multiplicity bounds). *Suppose that for a fixed (p, r) each nonramified factor of $\Phi_r(p)$ has exponent at most three and at most one has exponent three. If the pair is acceptable, then $\Phi_r(p) \leq rB^2Q_B(r)^2$. If instead at most one nonramified factor is squared, and none is cubed, then $\Phi_r(p) \leq rBQ_B(r)$. With at most one such squared factor and with its exponent at most three, the bound is $rB^2Q_B(r)$.*

Proof. Include the possible ramified factor r once. In the first case include every permitted nonramified prime twice, with one extra factor at most B ; the stated bound uses the harmless larger allowance B^2 . In the second case include every permitted prime once and the one extra factor at most B . A cubic factor requires at most two extra factors. Omitted factors only decrease these upper bounds. $\square$

Thus a strict inequality between the corresponding upper bound and $p^{r-1} < \Phi_r(p)$ excludes acceptability. These are integer comparisons, not floating-point logarithm estimates. Let $m = \lfloor (B-1)/(2r) \rfloor$. There are at most m possible factors in $Q_B(r)$, whence

$$rB^2Q_B(r)^2 \leq B^{2m+3}, \quad rBQ_B(r) \leq B^{m+2} \quad (r \leq B). \quad (2)$$

For a fixed base lower bound $b > 1$, the left sides of the comparisons $B^{2m+3} < b^{r-1}$ and $B^{m+2} < b^{r-1}$ are nonincreasing with r , while the right sides increase. A successful comparison at a tail threshold therefore covers all larger relevant indices. A finite exact check is needed below that threshold; checking only one index using $Q_B(r)$ would be invalid because that product is not monotone.

3.1 Recorded finite coverage at the candidate bound

The following is an account of the *recorded* coverage, subject to the audit limitations, rather than a replacement for the underlying calculations.

Range	Intended certificate and coverage
$r \geq 47,000$	Subgroups modulo ℓ^2 , no fourth-power hits, and at most one cubic factor per (p, r) ; exact squared-product cutoff for $p \geq 101$.
$30,594 \leq r \leq 46,999$	No distinct squared-factor collision; three cubic exceptions; linear bound excluding $p \geq 101$ outside checked exceptions.
$9,500 \leq r \leq 30,593$	Same collision condition; 28 cubic exceptions; base-stratified linear bounds.
$7 \leq r \leq 30,593$	All 3,299 odd prime indices: direct supported-factor sieves over every base not excluded analytically.
$p < 100$	All 24 odd prime bases: exact window $30,594 \leq r \leq 90,203$; base-specific squared-product cutoffs thereafter.
$r = 3, 5$	Direct complete polynomial sieves on all odd primes $p \leq B$.

The high-index squared cutoff checks 8,482 prime indices from 47,000 through 143,835, and (2) covers the tail beginning 143,836 with $b = 100$. The small-base squared-product check begins at 64,000, finds maximum last failure 90,203 (base 3), and uses coarse tail 294,499. The sharp linear threshold for prime bases at least 101 is 30,594. The direct small-base window reports 485,260,848 modular pair tests and no acceptable component. A noninteger-prime-base threshold of 100 is merely a convenient lower bound for primes at least 101.

For the middle range, the base cap is B below index 14,420, and successively 999,999,999, 99,999,999, 9,999,999, 999,999, 99,999, 9,999, and 999 at indices 14,420, 15,272, 16,340, 17,684, 19,458, 21,650, and 24,954. These cutoffs all lie within the collision-controlled range. Cubic exceptions above the applicable base cap are checked separately using $rB^2Q_B(r) < p^{r-1}$; those below it remain in the direct sieve. The 19 saved exception-bound comparisons are not silently absorbed into the generic square-only bound.

3.2 Enumeration and exact rejection

For a prime $\ell \equiv 1 \pmod{2r}$, choose a primitive generator g modulo ℓ^2 . Raising it to $\ell(\ell-1)/r$ gives an element h of order r . Its $r-1$ nonidentity powers enumerate all nontrivial r th roots modulo ℓ^2 . For every such root x , test all $p = x + t\ell^2$ with $3 \leq p \leq B$ prime. This enumerates precisely the square divisibilities with nonramified factor ℓ ; exact modular powers modulo ℓ^3 and ℓ^4 identify larger multiplicities. When $\ell^2 > B$ there is at most one candidate per residue. Smaller factors in the medium range are processed on the CPU with all lifts, not dropped by the GPU's size restriction.

The lower-index sieve instead enumerates the nonidentity roots modulo ℓ and their progressions of possible bases. It adds the ramified progression $p \equiv 1 \pmod{r}$ once. Every recorded factor is divided to its exact multiplicity. Large integer reconstruction is avoided when a safe upper-bit test already excludes smoothness. If the supported part is $A = \prod \ell^{e_\ell}$, put $U = \sum e_\ell \lceil \log_2(\ell+1) \rceil$. For odd primes this is the usual integer bit length, and $A < 2^U$. Consequently

$$U \leq (r-1) \lfloor \log_2 p \rfloor \implies A < p^{r-1} < \Phi_r(p).$$

All surviving cases are reconstructed with GMP and accepted only when division by their supported factors leaves one. Primality and exact reconstruction of every stored positive case are checked again by the separate Python validator. This establishes validity of listed cases, not absence of unlisted cases.

Computational obligation 3.2. At $B = 4,294,967,291$, the sieves and multiplicity/cutoff calculations exhaust all acceptable pairs. Every pair has $r \leq 47$; the complete catalog with $r \geq 7$ contains 12,553 pairs. The $r = 3, 5$ graph edges are also complete and exact.

The stored counts for $r = 7, 11, 13, 17, 19, 23, 29, 31, 47$ are respectively 12,496, 30, 14, 5, 3, 1, 2, 1, 1; all other retained indices have zero stored cases. The positive count was independently replayed. The completeness part of Obligation 3.2 remains subject to Section 7.

4 The rooted-hypergraph reduction

Let V be all odd primes at most B , and let $\mathcal{E}(p)$ contain every dependency set of an acceptable pair with base p . Define

$$T(X) = \{p \in V : \text{some } D \in \mathcal{E}(p) \text{ has } D \subseteq X\}.$$

Starting with V and repeatedly applying T gives a decreasing sequence. Its limiting set C is the ordinary component core.

Lemma 4.1 (Empty-core consequence). *If the catalog is complete and $C = \emptyset$, no odd perfect number supported in V can have Euler exponent greater than one.*

Proof. By Lemma 2.2, its nonempty support S would have $S \subseteq T(S)$. Induction gives $S \subseteq T^j(V)$ for all j , contradicting the empty limiting core. $\square$

For Euler exponent one, write $D_q = \text{supp}((q+1)/2)$. An actual support S containing q must satisfy $D_q \subseteq S$ and every $p \in S \setminus \{q\}$ must have some edge contained in S . Here D_q contains only primes below q , which are ordinary and therefore initially viable.

Lemma 4.2 (Candidate-root filters). *If $C = \emptyset$, an actual Euler root q has $q \equiv 1 \pmod{4}$, every member of D_q has at least one edge, and q occurs in some catalog edge.*

Proof. Only the last assertion needs further explanation. If no selected edge of any ordinary member of S contained q , then $S \setminus \{q\}$ would be a nonempty postfix set for T , contradicting emptiness of C . This set is nonempty since a single odd prime power cannot be perfect: $\sigma(q^a)/q^a < q/(q-1) < 2$. Thus some selected prime-index edge contains q . This reasoning, rather than $q \mid \sigma(N)$ alone, justifies the filter; a prime can divide a composite-index cyclotomic factor without dividing a prime-index factor of the same source. $\square$

Give each vertex its synchronous deletion level $L(p)$, with level zero for vertices having no edges and level $j \geq 1$ for those removed at iteration j starting from the initially viable set. When the core is empty every level is finite. Every edge at a positive-level vertex has a dependency of strictly lower level. Otherwise all its dependencies would still have been alive when that vertex was deleted.

4.1 Simultaneous labels and their soundness

Let Q be the candidate roots passing Lemma 4.2. In ascending level order set

$$R(p) = (\{p\} \cap Q) \cup \bigcup_{D \in \mathcal{E}(p)} \bigcap_{\substack{t \in D \\ L(t) < L(p)}} R(t). \quad (3)$$

The union is empty for level-zero vertices, and every intersection displayed at positive level is over a nonempty set. Then repeatedly delete $q \in R(p)$ for $p \neq q$ whenever no edge at p has q in every dependency's label set. Keep a candidate's own label at that vertex. Finally retain q only if $q \in R(t)$ for every $t \in D_q$.

Lemma 4.3 (Conservative labels). *For every rooted support (S, q) satisfying the preceding necessary conditions, $q \in R(p)$ for every $p \in S$, both initially and throughout refinement. Hence rejecting every final candidate rejects every such support.*

Proof. Induct on the deletion level of a vertex in S . A level-zero ordinary vertex cannot occur because it has no edge; the root retains its own label. For an ordinary p , choose an edge $D \subseteq S$. All its lower-level dependencies have label q by induction, so their intersection, and thus the union in (3), contains q . For refinement, suppose the invariant holds immediately before a deletion. An ordinary vertex in S has its chosen edge inside S , so every dependency still contains q ; its true label cannot be deleted. This argument also justifies in-place, rather than simultaneous, refinement. At the end every member of $D_q \subseteq S$ still has q and the root passes the last test. $\square$

Computational obligation 4.4. For the complete graph in Obligation 3.2, the ordinary core is empty and the final rooted-label test retains no candidate Euler root.

Theorem 4.5 (Conditional candidate bound). *If Obligations 3.2 and 4.4 hold at $B = 4,294,967,291$, every odd perfect number N has $P(N) > B$.*

Proof. Suppose $P(N) \leq B$. If its Euler exponent exceeds one, Lemma 4.1 contradicts Obligation 4.4. Otherwise Lemmas 4.2 and 4.3 show that its Euler root survives, again contradicting that obligation. $\square$

The recorded graph has 203,280,220 vertices and becomes empty after ten ordinary-deletion rounds. There are 11,859,489 candidate roots, 48,395,333 initial label memberships, and 11,859,734 final memberships after five refinement rounds; none passes the final root test. The Rust and C++ summary lines agree. Individual deletion levels, edges and label sets were not retained as a replay certificate.

4.2 Algorithm outline

```

Enumerate  $V$  and every permissible prime index  $r$ .
Exclude analytically covered (base,index) ranges.
For each remaining  $r$ , generate all supported factors and exact valuations.
Reject by the safe bit bound or by a nonunit exact cofactor.
Construct every accepted dependency set  $E(p)$ .
Synchronously remove vertices with no edge in the live set.
If a nonempty core remains: return unresolved.
Compute candidate Euler roots and ascending-level labels.
Refine labels using all dependencies until no deletion occurs.
Return candidates whose Euler-component dependencies retain their label.

```

All loops are finite under Lemma 2.3. A resource limit, incomplete checkpoint collection, surviving root, or unhandled multiplicity must return unresolved, never a successful proof.

5 Computational methods and verification architecture

The production arithmetic is Rust with a GMP interface; a separately written C++20 implementation reproduces the main kernels. Metal and CUDA scan subgroup residues. The CUDA file incorporates the C++ host implementation, so it is not independent of that code. Both native implementations use the same mathematical classification, related sieve strategies, and GMP. Independence here means separate implementations with comparison checks, not statistically independent errors, independent mathematical assumptions, formal verification, or human peer review.

The workload benefits from segmented prime sieving, packed odd-prime masks with rank lookup, compact factor links, exact modular arithmetic, Montgomery reduction, loop unrolling and batched GPU execution. The audit does not claim that custom allocators, a new assembly implementation or a globally optimal language choice were established. Rust won eight of nine *small* rerun workloads; the GMP case was essentially tied. These microbenchmarks support the existing implementation choice only at their measured scale.

Local quick workload	Rust median (ms)	C++ median (ms)
Prime sieve to 10^6	0.918	1.684
25,000 modular exponentiations	2.552	5.314
5,000 GMP operations	2.051	2.047
Φ_3 sieve to 25,000	0.439	0.663
Φ_5 sieve to 25,000	0.288	0.451
Dependency closure to 20,000	0.561	0.875
Abundance expression to 20,000	0.452	0.831
Subgroup scan to 30,000, indices 7–47	2.072	3.499
Four-thread prime sieve to 10^6	0.353	0.575

The medians are three alternating-order repetitions on an Apple M4 Max, 16 CPU cores, 128 GiB memory. The machine was not isolated from other work; submillisecond differences should not be extrapolated. Exact outputs, not just times, were compared. The whole invocation, including compilation and its positive control, took 229.039 seconds. Historical timings include 1,932.959 seconds (Rust) and 2,120.432 seconds (C++) for the principal root calculation, and 33,452.979 seconds summed across the 100 high-index GPU shards. The latter is accumulated kernel time, not elapsed time or a bill; the archive does not give complete per-shard hardware or compiler provenance.

5.1 Checks rerun in the 26 August audit

The independent artifact audit reconstructed all 12,553 listed values and checked 71,608 distinct primality inputs by trial division, including the bound and next prime. It verified 33 stored cubic witnesses and three coarse tail inequalities. It compared 26 cached Rust/C++ result pairs and the 20 source digests in the saved principal summary; 81 source references across 13 supplementary certificates also matched. None of these counts implies exhaustiveness of a negative search.

There were 328 passing Python tests and 116 passing Rust tests. The abstract root-label tests include all 32,768 three-vertex configurations and 25,000 seeded random hypergraphs, with positive controls. These support the implementation; the general soundness argument is Lemma 4.3, not an inference from those samples.

An independent direct-factorization program regenerated the *complete* graph for $B = 1000$: 167 odd primes, all 94 permissible odd prime indices, 67 accepted components, an empty ordinary core and no surviving Euler root. It checks all 80 primes $q \equiv 1 \pmod{4}$ separately by fixed-root closure; it does not use subgroup sieving or root labels. This is a small known bound, not a new result. Both native programs agreed on zero survivors.

The historical complete command was rerun in a disposable copy after building the native binaries. It passed in 75.526 seconds, including a fresh aggregate subgroup-work count. Its cached parts remained cached. An initial attempt without the required binary failed and is retained in the logs rather than erased. Five full-bound cutoff/window workloads were freshly recomputed in both Rust and C++; all matched the historical non-timing fields. Their individual completion records and exact comparison outcomes are in `audit/native-recomputations/` and `COMPUTATION.md`. All 19 saved cubic-exception inequalities were also recomputed in both languages. A restricted actual Metal scan matched Rust and C++ on 1,558 subgroup steps,

16 square hits and one known cubic hit. No large GPU exclusion scan or full B -scale root graph was regenerated.

6 Full-system necessary conditions beyond a fixed prime bound

The following arguments summarize selected later work through 2 September 2026, reconstructed for this revision on 3 September. They do not assume the numerical obligations of Theorem 4.5, and they do not discharge them. All conclusions concern an *entire* hypothetical odd perfect number, not a selected dependency tree. Their additional hypotheses are stated explicitly. No novelty or external-validation claim is made.

6.1 Complete rows, valuations, and the Euler column

Write the Euler exponent as $\alpha := a$, set $a_q = \alpha$, and put

$$S = O \cup \{q\}, \quad n_x = a_x + 1, \quad G_n(X) = \sum_{j=0}^{n-1} X^j, \quad V_x = \frac{G_{n_x}(x)}{2^{\lfloor x=q \rfloor}}, \quad b_{xr} = \nu_r(V_x).$$

Here O is the ordinary prime support. These actual odd rows satisfy

$$b_{xx} = 0, \quad V_x = \prod_{r \in S} r^{b_{xr}}, \quad \sum_{x \in S} b_{xr} = a_r, \quad \prod_{x \in S} V_x = N. \quad (4)$$

Every output, repeated valuation and outside input relative to any chosen subset is retained. There are at least two support primes, since a singleton support contradicts a positive column and the zero diagonal. The support and exponents are otherwise arbitrary.

An ordered physical edge $x \rightarrow r$ means $b_{xr} > 0$. Let W count those edges with $b_{rx} = 0$, including edges of even value. Such an edge can lie on a longer cycle. *Full reciprocity* is the additional hypothesis $W = 0$, not a property assumed of all odd perfect numbers. For $x \in O$ put $d_x = \text{ord}_q(x)$ and $h_x = \nu_q(n_x)$. Geometric LTE gives

$$\alpha = I_q + T_q, \quad I_q = \sum_{\substack{x \in O, d_x > 1 \\ d_x | n_x}} \nu_q(x^{d_x} - 1), \quad T_q = \sum_{\substack{x \in O \\ d_x | n_x}} h_x. \quad (5)$$

At a raw source $d_x = 1$, the initial valuation cancels and the full input is h_x . Ineligible indices contribute nothing. These are parts of the same column, not independent resources.

For $\alpha > 1$, let $\beta = (\alpha + 1)/2$, an odd integer greater than one. The Euler row splits into coprime odd factors

$$V_q = P_+ P_-, \quad P_+ = G_\beta(q) > 1, \quad P_- = (q^\beta + 1)/2 > 1.$$

Their coprimality follows by subtracting $q^\beta - 1$ from $q^\beta + 1$. Put $\mathcal{B} = \text{supp}(\beta)$, $k = |\mathcal{B}|$, and

$$\mathcal{G} = \{r \in O : r \mid P_-, \lambda_q(r) = 1\}, \quad \lambda_q(r) = \nu_q(r^{\text{ord}_q(r)} - 1).$$

Every such head is ordinary and distinct from q , even if $q \mid \beta$.

Lemma 6.1 (Subset allocation). *For any $H \subseteq \mathcal{G}$, let $T_H = \sum_{r \in H, d_r | n_r} h_r$. Then $W + T_H \geq |H|$.*

Proof. The full return at $r \in H$ is h_r if raw, $1+h_r$ if nonraw eligible, and zero otherwise. A zero return gives the one-way edge $q \rightarrow r$. If the return is positive odd, put $o_r = (r-1)/2^{\nu_2(r-1)}$. Since $r \mid P_-$, the order of q modulo r has two-part exactly two; thus $q^{o_r} = -1 \pmod{r}$. The entire ordinary row, reduced modulo r and raised to o_r , gives

$$\prod_{\substack{z \in O \\ b_{rz} > 0}} z^{b_{rz} o_r} = -1 \pmod{r}.$$

If every such output satisfied $\nu_2(\text{ord}_r(z)) \leq \nu_2(b_{rz})$, each would equal one, a contradiction. Some actual ordinary output z consequently has even $\text{ord}_r(z)$, which cannot divide the odd n_z . Hence $z \rightarrow r$ is absent, and $r \rightarrow z$ is one-way. Choose one for each such r . The chosen ordinary-source edges have distinct sources and are distinct from the direct q -source edges, even if receivers coincide. Finally every positive even return consumes $h_r \geq 1$ when nonraw, or $h_r \geq 2$ when raw, in its own summand of T_H . This proves the count without replacing any full coefficient by a unit. $\square$

Theorem 6.2 (Whole-system edge/index alternative). *If $\alpha > 1$, then*

$$W + T_q \geq 2^k - k. \quad (6)$$

If $\alpha = 1$, then $W \geq 1$. In particular full reciprocity forces $\alpha > 1$ and $q \mid n_x$ for some ordinary x .

Proof. For a q -unit u , set $F_q(u) = (u^{q-1} - 1)/q \pmod{q}$. It is additive on products, and $F_q(r) \neq 0$ exactly when $\lambda_q(r) = 1$. The cyclotomic product gives $\Phi_j(X) \equiv 1 - \mu(j)X \pmod{X^2}$ for $j > 1$. Thus for every squarefree $d > 1$ dividing β ,

$$F_q(\Phi_{2d}(q)) = -\mu(d) \neq 0.$$

Each of the $2^k - 1$ layers therefore contains a member of $\mathcal{G}$. Let $\mathcal{G}_{\text{sf}}$ be the members occurring in these layers.

For an odd prime $r \mid P_-$, write $\text{ord}_r(q) = 2e$ with e odd. Its full initial valuation occurs in $\Phi_{2e}(q)$; subsequent occurrences are exactly one copy in each $\Phi_{2er^j}(q)$, $j \geq 1$. Indeed summing these values over divisors of an index gives precisely the LTE valuation of the corresponding power minus one, and divisor induction identifies each cyclotomic valuation. Also $2e \mid r-1$. Among squarefree $d > 1$, the head can therefore occur only at $d = e$ and $d = er$. Double occurrence requires $r \in \mathcal{B}$, $e > 1$, and all prime divisors of e smaller than r . The least prime in $\mathcal{B}$ cannot double. At most $k-1$ heads double, giving

$$|\mathcal{G}_{\text{sf}}| \geq (2^k - 1) - (k - 1) = 2^k - k. \quad (7)$$

Apply Lemma 6.1 and $T_{\mathcal{G}} \leq T_q$. This retains initial/ramified overlaps rather than assuming coprime layers.

For $\alpha = 1$, the complete q column has exactly one unit payer x . If $b_{qx} = 0$, then $x \rightarrow q$ is one-way. Otherwise x divides $(q+1)/2$, and the odd-return argument in the preceding lemma again produces an ordinary one-way edge; it needs no initial-depth assumption. Thus $W \geq 1$. In the reciprocal branch, (6) gives $T_q > 0$, proving the final assertion. $\square$

Theorem 6.3 (Localized compensation for a powerful half-index). *Suppose $\alpha > 1$ and every prime divisor of β occurs to exponent at least two. Set*

$$T_{\text{out}} = \sum_{\substack{x \in O \setminus \mathcal{B} \\ d_x \mid n_x}} \nu_q(n_x).$$

Then $|\mathcal{G} \setminus \mathcal{B}| \geq 2^k - k$ and

$$W + T_{\text{out}} \geq 2^k - k. \quad (8)$$

In particular $W = 0$ forces this index multiplicity outside $\mathcal{B}$.

Proof. For each $\ell \in \mathcal{G} \cap \mathcal{B}$, write $\text{ord}_\ell(q) = 2e_\ell$. The layer with half-index $e_\ell \ell^2$ is available because $e_\ell \mid \beta$, $\ell \nmid e_\ell$, and $\ell^2 \mid \beta$. Its complete factorization has the shape

$$\Phi_{2e_\ell \ell^2}(q) = \ell \prod_z z^{c_z}, \quad \text{ord}_z(q) = 2e_\ell \ell^2.$$

The ramified factor has value exactly one by the preceding valuation rule. Every other factor is primitive: a nonprimitive factor at this index would have to be its largest prime factor, which is ℓ since $2e_\ell \mid \ell - 1$. The layer's Fermat quotient is zero, while $F_q(\ell) \neq 0$. Consequently at least one primitive head z has $c_z F_q(z) \neq 0$ and belongs to $\mathcal{G}$. Its original half-order is nonsquarefree, so it never occurs in the squarefree pool $\mathcal{G}_{\text{sf}}$. Heads chosen for different ℓ are distinct: their exact half-order recovers ℓ as its largest prime factor. Therefore

$$|\mathcal{G}| \geq |\mathcal{G}_{\text{sf}}| + |\mathcal{G} \cap \mathcal{B}|, \quad |\mathcal{G} \setminus \mathcal{B}| \geq 2^k - k.$$

Apply Lemma 6.1 to $H = \mathcal{G} \setminus \mathcal{B}$; its T_H is a subsum of T_{out} . The layer value c_z was used only to find a simple head; its full Euler input may be $c_z + \nu_z(\beta)$ and was not assumed prime to q . $\square$

Neither powerful β nor an incompatible upper bound on W or T_{out} has been proved universal. Index multiplicity need not come from distinct sources. Already (7) gives $|\mathcal{G} \setminus \mathcal{B}| \geq 2^k - 2k$; the localized theorem is a sharper conditional count, not the first observation of an outside head when $k \geq 3$. Repeated-layer refinements, exact first-power resonance and the failure of aggregate higher digits to force individual depth are recorded in the supplementary update, not used as unstated hypotheses.

6.2 A forced seed at an all-raw supported-index frontier

Proposition 6.4 (Weighted gap and one-step exit). *Assume full reciprocity. Let*

$$L = \max\{p \in S : p \mid n_x \text{ for some } x \in S\}, \quad B_L = \{x : b_{Lx} > 0\}.$$

If every $x \in B_L$ satisfies $x \equiv 1 \pmod{L}$, then $L > q$. With $f_x = \nu_L(x - 1)$, $h_x = \nu_L(n_x)$ and $c_x = b_{Lx}$,

$$\sum_{x \in B_L} f_x c_x \leq a_L - 1 = \sum_{x \in B_L} h_x - 1. \quad (9)$$

The set $A_L = \{x \in B_L : h_x > f_x\}$ is nonempty. For a member x_ maximizing f_x on A_L , put $f_* := f_{x_*}$. Every prime head of the actual row factor*

$$K_* = \Phi_{L f_* + 1}(x_*)/L$$

is outside A_L . Such a head is either a passive neighbor with $1 \leq h_z \leq f_z$, or a nonneighbor of L with $h_z = 0$.

Proof. Theorem 6.2 gives $\alpha > 1$ and puts q in the finite set defining L , so $L \geq q$. Larger absent index primes remain possible. For any supported $z > L$, maximality gives $\nu_z(n_y) = 0$ for every source y . Every positive input at z is therefore nonraw and initial-only. Reciprocity makes B_L the full set of suppliers of L as well as its output set. If $L = q$, the raw-neighbor hypothesis makes $V_q \equiv 1$ modulo q , contrary to $V_q \equiv 1/2$. Thus $L > q$, and all its raw

neighbors satisfy $x > L$ and are ordinary. Raw LTE and the full column give $b_{xL} = h_x \geq 1$ and $a_L = \sum h_x$. Also $x \geq 2L^{f_x} + 1$ since both primes are odd. Hence, for $M = \sum f_x c_x$,

$$2L^M < \prod_{x \in B_L} x^{c_x} = G_{a_L+1}(L) < \frac{L^{a_L}}{1 - 1/L} < 2L^{a_L}.$$

Integrality proves (9); some $h_x > f_x c_x \geq f_x$, so A_L is nonempty.

For the stated seed, $L^{f_*+1} \mid n_{x_*}$. LTE removes exactly one L from $\Phi_{L^{f_*+1}}(x_*) = G_L(x_*^{L^{f_*}})$, leaving an odd integer $K_* > 1$ dividing its actual row. For every prime $z \mid K_*$,

$$\text{ord}_z(x_*) = L^{f_*+1}, \quad z \geq 2L^{f_*+1} + 1, \quad f_z \geq f_* + 1.$$

Indeed the geometric sum forces $\text{ord}_z(x_*^{L^{f_*}}) = L$, and the order-quotient formula gives the asserted exact order. Such z is supported, ordinary and above L . Its full forward value is exactly $b_{x_*z} = \nu_z(K_*)$, since no receiving-prime index lift is available. Maximality of f_* excludes z from A_L . If $z \in B_L$, it is passive; otherwise reciprocity gives $b_{zL} = 0$, and raw LTE makes this equivalent to $h_z = 0$. $\square$

The full coefficient in this proposition need not be one, odd, or small. Every head still has its physical reverse to x_* , and later re-entry is possible. Neither exit case supplies the next divisor $L^{f_z+1} \mid n_z$. If the all-row hypothesis fails, the complete L column also contains nonraw initial valuations; these cannot be deleted from the weighted comparison. An Euler supplier can have even eligible order (two-part two). The frontier is the largest *supported index prime*, not the largest prime label or the largest prime divisor of the product of all indices. These unexcluded cases prevent a universal descent conclusion.

6.3 Collision mass and an imported unequal-index estimate

Set $s = |S|$, $m = \min S$, $d_q = |\text{supp}(V_q)|$, and define the residual integer

$$\mathcal{R} = \frac{N}{\text{lcm}_{x \in S} V_x} = \prod_{r \in S} r^{\delta_r}, \quad \delta_r = a_r - \max_x b_{xr}.$$

This is not the ordinary component core of the earlier finite graph. For positive entries $c_1 \geq \dots \geq c_t$ in one receiving column, $\delta_r = \sum_{j=2}^t c_j$, while its exponent in the product of unordered pairwise row gcds is $\sum_{j=2}^t (j-1)c_j$. Thus

$$\mathcal{R} \mid \prod_{x < y} \gcd(V_x, V_y), \quad \log \mathcal{R} \leq \sum_{x < y} \log \gcd(V_x, V_y). \quad (10)$$

Pairwise gcds overcount columns with three or more suppliers; they are not independent budgets. Since every row exceeds one, counting physical edges and using $\delta_r \geq t_r - 1$, where t_r is the number of positive suppliers of r , gives

$$\log \mathcal{R} \geq (d_q - 1) \log m. \quad (11)$$

When $\alpha > 1$, the coprime Euler halves ensure $d_q \geq 2$. Under full reciprocity, every negative Euler head has positive even full return, by the odd-return gate above. Let $h = \omega(P_-)$. The odd total α also requires a positive-side odd payer. After removing just the largest q -column input, at least $2h - 1$ units remain. Consequently the stronger floor is

$$\log \mathcal{R} \geq F := \max\{(d_q - 1) \log m, (2h - 1) \log q\}. \quad (12)$$

The maximum, not a sum, avoids charging the column twice. Without reciprocity we use $F = (d_q - 1) \log m$ instead.

Corvaja–Zannier’s Corollary 1 gives the following specialization for fixed distinct primes x, y and $\epsilon > 0$: outside a finite set $E_{x,y,\epsilon}$ of positive exponent pairs,

$$\log \gcd(x^u - 1, y^v - 1) < \epsilon \max(u \log x, v \log y). \quad (13)$$

Apply their fixed group generated by $(x, 1), (1, y)$; positive prime powers are multiplicatively independent. Their finite-place sum is minus the logarithm of the integer gcd, and the real-place term is zero. This imports the stated deep theorem, not a proof of it [1, Corollary 1, equation (1.3)].

Proposition 6.5 (Conditional same-total gcd test). *If every actual pair (n_x, n_y) avoids $E_{x,y,\epsilon}$, then*

$$\log \mathcal{R} < \epsilon \sum_{x < y} \max(n_x \log x, n_y \log y) \leq \epsilon(s-1)(\log N + \log \text{rad } N). \quad (14)$$

Such data are impossible if the first upper expression is also at most the applicable floor F .

Proof. Each V_x divides $x^{n_x} - 1$, including the single Euler normalization. Apply (13) at the original indices to every pair, then use (10). There is at least one pair, so strictness is preserved. Finally $\max(u, v) \leq u + v$, each source appears $s-1$ times, and $\sum_x n_x \log x = \log N + \log \text{rad } N$. The final assertion contradicts the lower floor for the same integer. $\square$

No least-common-multiple inflation of the two indices is needed here. However, the exceptional sets depend on the prime pair and ϵ ; choosing a smaller ϵ does not prove their avoidance. Support size and bases vary, and the guaranteed collision floors need not be a fixed fraction of $\log N$. No effective uniform exception list or adequate opposing bound has been obtained. The gcd test consequently excludes no new complete class without its explicit additional conditions.

6.4 What is still needed for a universal proof

Theorems 6.2 and 6.3 do not bound their left sides incompatibly. Proposition 6.4 does not handle nonraw neighborhoods or continue past passive/exterior exits. Proposition 6.5 does not control exceptional pairs or force sufficient collision mass. None establishes that a previously studied fan, cycle, norm, equal-weight, or special-tree hypothesis occurs universally. The primary open obligation remains a contradiction covering every support and exponent pattern, or a fully verified example.

7 Audit gaps and limits

GPU counter safety. Both GPU kernels increment a 32-bit hit counter. Once a batch finishes, the host rejects a count larger than the hit buffer capacity, normally 2,000,000. But $2^{32} + 17$ increments leave the counter equal to 17, so that guard is not generally fail-closed. A configured batch can have 20,000,000 jobs of 8,192 iterations, which exceeds 2^{32} . No sticky overflow flag or recorded proof of a tighter per-batch hit bound was found. The mathematical control for this defect is not evidence that an actual historical batch overflowed. Nevertheless the summaries alone do not exclude that possibility. Resolution requires a valid bound for the exact batches or regeneration with a provably safe counter/overflow mechanism. Small samples and ordinary observed hit counts do not discharge this obligation. The medium driver defaults to only 200,000 jobs per batch, giving a nonwrapping iteration bound; the high-index allegation is not extended to that safe default configuration. A future high-index run can use 500,000 jobs per batch: $500,000 \cdot 8192 = 4,096,000,000 < 2^{32}$. For whole-group Metal batching, each group’s job count is also below that cap at the present index ceiling. This addresses counter wrap for that future configuration, not the historical record.

Negative certificates. Positive factorization records are short, exactly checkable witnesses. A summary reporting no acceptable omitted values, no fourth powers, or no root survivors is different. The retained medium-index files omit the full stream of square events used for collision detection. The high-index shards omit the exhaustive residue stream, and the root cache omits a per-vertex proof object. Checksums help detect accidental discrepancies but do not prove the absence of unrecorded events. Full independent recomputation, or richer replayable exclusion and deletion certificates, remains necessary for the stronger certification requested for this work.

Arithmetic and provenance. For $p < 2^{32}$, $\Phi_5(p) < (p+1)^4 \leq 2^{128}$, and prime squares fit unsigned 64-bit words. Squares near B^2 exceed the signed 64-bit range, so the Montgomery carry cannot be discarded; the source handles it explicitly. The Rust GMP interface assumes a 64-bit Unix ABI and must not be ported unchanged to an incompatible ABI. Historical sources now match the final summary’s digests, but those digests were taken at replay time and are not per-run binary attestations. That distinction is preserved in the manifest.

Abundance and secondary research. The historical abundance program evaluates a restricted prime-set expression. Its numerical inequality is not by itself a universal contradiction without the hypotheses that put the actual support in that set. We do not use it in Theorem 4.5. The supplement records its exact interpretation. The separate 25-distinct-prime result assumes Euler component 157^1 and two specified complete-component walks. It is not a bound for all odd perfect numbers. Experiments and countermodels there do not settle unrestricted exponent closure or nonexistence.

8 Reproducibility and availability

The versioned local review bundle contains the LaTeX source, bibliography, PDF, Markdown companion, audit, supplement, claim-to-evidence table, original source snapshot, all saved `target/record-proofs/` checkpoints, supplementary certificates, sanitized model provenance, and new run logs. A manifest lists byte counts and SHA-256 digests; the external archive has its own checksum. Historical notebooks are retained unchanged, including superseded optimistic headings. The audit controls their interpretation.

`REPRODUCE.md` distinguishes (i) offline positive-witness/checkpoint replay, (ii) independent recomputation of specified finite tasks, and (iii) full regeneration. It records what each checker trusts. No authentication or cloud account is needed for replay. Legacy cloud orchestration is retained as historical source, not invoked by the reproduction entry points. The finalisation inventory found no tagged research EC2 instances, EBS volumes, allocated addresses or spot requests in 17 enabled regions, and no labeled Daytona workers. Four small historical S3 objects were copied locally and the bucket was preserved. Untagged or unrelated resources were not changed.

No public repository, DOI deposit, submission or reviewer contact is represented as having occurred. The package is for the author’s review and then, only with approval, external scrutiny. A public code/data license and any venue-specific declarations require author decisions. No historical cost total is asserted.

The 3 September revision adds the later proof notes and dated internal reconstructions, with a change log in `LATEST_RESEARCH.md`. The 26 August sealed export is preserved. Compilation, PDF inspection and file-integrity checks for the revision are documented separately from historical mathematical runs. No new numerical certificate, exhaustive negative-search replay or full candidate graph regeneration is implied.

9 Author contributions and AI assistance

Richie McIlroy selected and directed the project, set its record-oriented and computational objectives, made scope and resource decisions, requested stronger verification, and instructed finalisation with honest disclosure. The available record does *not* establish that he personally checked the mathematics or code. His name on this requested draft is not a statement that final approval or submission approval has been given.

AI assistance was substantive research, not merely copy-editing. AI systems proposed mathematical ideas and failed approaches, drafted proofs, implemented native and interpreted programs, designed and operated experiments, wrote validators, compared outputs, performed the present audit, and drafted this paper and supplementary documentation. Automated checks were run by the AI agent. They are not a recorded independent human review of either the original research or this audit.

The AI assistance is referred to in this draft as **GPT-6 Astra**. Historical model-version details are not fully verified; phase-specific run metadata are preserved separately rather than replaced by this name. Available records identify OpenAI and the Codex Desktop interface for the initial work. They distinguish candidate-bound development on 25–26 August, re-examination and finalisation on 26 August, later theoretical research through 2 September, and this revision on 3 September. Complete worker-level model mappings, immutable versions and per-artifact attribution are not established. No current setting is used to infer a historical model identity.

The later written checks used distinct AI roles with shared context; they are not independent human or formal verification. No new mathematical program run was performed for the September revision. The bundle retains the metadata extracts, not raw private conversations or credentials. No AI system is listed as a co-author.

The FunSearch paper explicitly identifies the generative model, evaluation process and available code [8]; Georgiev and co-authors distinguish AI construction, evaluation, proof-related work and limitations [2]. These are examples of transparent reporting, not endorsements of this result. The LMS policy requires disclosure extending to ideas, calculations and proof construction/verification, and human checking before submission [5]. Nature Portfolio likewise retains human accountability and requires transparent use [7]. This disclosure uses policy descriptions recorded in the 26 August audit; fresh retrieval of those two policy pages failed during this revision. Their current text must therefore be checked again before any submission. This disclosure does not substitute for that human work or guarantee compliance with a venue not yet selected. The entire draft remains subject to McIlroy’s review and approval; no affiliation, qualifications, funding arrangement, collaborators or author approvals have been invented.

10 Conclusion

The finite reduction is explicit and the saved evidence is substantial, but the principal numerical statement remains a candidate under this audit. The later full-system arguments give necessary constraints and precisely delimited conditional refinements, not a universal contradiction. The next review steps are finite: verify both sets of arguments, resolve GPU counter safety, assess or regenerate the negative-search and graph evidence, check priority and obtain accountable human review. This revised package records the current evidence and gaps; it does not complete the existence problem.

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